

Robot Camera System Documentation

This document provides a comprehensive guide for setting up, configuring, and operating the robot camera system.

The system integrates a Raspberry Pi 5 with a Raspberry Pi Camera Module V2 and an MLX90640 thermal camera module. It enables real-time streaming of camera feeds via a web interface.

Hardware Components

- Raspberry Pi 5
- Raspberry Pi Camera Module V2
- MLX90640 Thermal Camera Module (32x24 IR sensor array, I2C interface).
- Jio wifi

Hardware Connections

- Connect Rpi camera with the ribbon cable

- MLX Camera:

MLX90640 Pin	Raspberry Pi 5 Pin
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VCC	3.3V (Pin 1)
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GND	GND (Pin 6)
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SDA	GPIO 2 (Pin 3)
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SCL	GPIO 3 (Pin 5)
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- Test Rpi camera with [`rpicam-hello`] in terminal
- Test I2C with [`i2cdetect -y 1`]; the MLX90640 should appear at address 0x33

Software Setup

1. Navigate to the project directory: camera

```
[ cd camera ]
```

2. Activate the virtual environment: camera_env

```
[ source camera_env/bin/activate ]
```

3. Start the Hosting Server (in the first terminal, with venv activated):

```
[ python3 -m http.server 8000 ]
```

4. Start the Camera Stream (in a second terminal, with venv activated):

```
[ python3 cobra_stream.py ]
```

5. Access the Dashboard:

- Find the Pi's IP address: `hostname -I` (e.g., 192.168.1.100).
- Open a browser on any device on the same network: [http://\[IP_ADDRESS\]:8000](http://[IP_ADDRESS]:8000)
(Example- `http://192.168.1.100:8000`).
- Click Connect to initiate streaming.

Required Dependencies

- Rpicam apps

```
[ sudo apt install rpicam-apps libcamera0 -y ]
```

- GStreamer

```
[ sudo apt install gstreamer1.0-tools gstreamer1.0-plugins-base gstreamer1.0-plugins-good  
gstreamer1.0-plugins-bad gstreamer1.0-plugins-ugly gstreamer1.0-libav gstreamer1.0-  
libcamera libgstreamer1.0-dev libgstreamer-plugins-base1.0-dev -y ]
```

- Open CV Build

```
[ sudo apt install build-essential cmake git pkg-config libgtk-3-dev libavcodec-dev  
libavformat-dev libswscale-dev libv4l-dev libxvidcore-dev libx264-dev libjpeg-dev libpng-  
dev libtiff-dev gfortran openexr libatlas-base-dev python3-dev python3-numpy libtbb2 libtbb-  
dev libdc1394-dev -y ]
```

Drone Documentation

Hardware Components

- RadioMaster Pocket Controller: ELRS transmitter
- BetaFPV 5-in-1 Flight Controller: An all-in-one board integrating FC, ESC, OSD, VTX, RX
- HappyModel SE0702 KV2 Motors:
- Whoop Frame
- 1S Battery

Binding the RC Controller to the Drone

- Power on your RadioMaster Pocket and ensure it's set to ELRS mode (EdgeTX menu: Model Setup > Internal RF > ELRS).
- Plug in the 1S battery to the drone three times in rapid succession (connect for 1-2 seconds each time, then disconnect). On the third connection, the drone's LED should flash rapidly, indicating bind mode.
- On the RadioMaster Pocket, go to the ELRS script, select "Bind," and confirm. The drone's LED should go solid or change pattern to indicate a successful bind.

Setting Up the Drone in BetaFlight Configurator

- Connect your drone via USB (use a data-capable cable plugged into the FC's USB port).
- Flashing Firmware Steps (if updating):

Open BetaFlight Configurator > Firmware Flasher tab.

Select your target (BETAFLPVG473) and version (4.5.0).

Click "Load Firmware [Online]" or "Load Firmware [Local]" > "Flash Firmware."

After flashing, connect and verify the version in the top bar.

Step-by-Step Configuration

1. Basic Connection and Sensor Alignment (Setup Tab)

- Click connect (select the Betaflight COM port if needed).
- The 3D model should appear; verify firmware version and sensors (Gyro must show).
- Place the drone on a flat surface. Click Calibrate Accelerometer to level it
- Hold the drone nose-forward and tilt/roll it—the 3D model must mirror movements exactly.

2. Receiver and RC Controls Setup

- This maps your RadioMaster Pocket's switches to drone channels for throttle, yaw, pitch, and roll.
- In Ports tab: Ensure Serial RX is enabled on the UART for your integrated ELRS RX
- In Receiver tab: Set Receiver Mode to Serial-based. Enable Telemetry for battery/RSSI feedback.
- Bind the receiver while connected.
- In Modes tab: Assign switches. Drag ranges for Angle Mode. Test by flipping switches—sliders should activate.
- Save and Reboot.

3. Motor Controls and Testing (Motors Tab)

- Verify spin direction, order, and protocol for SE0702 motors.
- In Configuration tab: Set ESC/Motor Protocol to DShot300
- In Motors tab: Use the master slider or individual motor sliders to spin each at low throttle and follow the motor order to set given motor directions.

Meta Quest 3S Documentation

1. Initial Setup and Power-On

- Power on the Meta Quest 3S headset.
- Verify that the hand trackers (probes) are fully charged.

2. Home Screen Applications

- Once on the home screen, locate the Unity builds that have been uploaded.
- These builds support simulation features and video streaming capabilities.